

Change the title of Clause 163 (added by IEEE Std 802.3ck-2022) as follows:

163. Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, and 800GBASE-KR8

163.1 Overview

Change the first three paragraphs in 163.1 as follows:

This clause specifies the 100GBASE-KR1-PMD, ~~the 200GBASE-KR2-PMD, the 400GBASE-KR4-PMD, and 800GBASE-KR8 PMDs~~ and the baseband medium. The specifications for the ~~three~~ four PMDs are similar, except for the number of lanes and associated parameters and the MDI.

There are two associated annexes. [Annex 163A](#) provides measurement methods and test points for backplane and chip-to-chip interfaces. [Annex 163B](#) provides information on an example test fixture.

When forming a complete Physical Layer, a PMD shall be connected as illustrated in [Figure 163–1](#), to the appropriate sublayers (as specified in [Table 163–1](#), [Table 163–2](#), ~~and Table 163–3~~, and [Table 163–3a](#)), to the medium through the appropriate MDI, and to the management functions that are optionally accessible through the management interface defined in Clause 45, or equivalent.

Insert Table 163–3a after Table 163–3 as follows:

Table 163–3a—Physical Layer clauses associated with the 800GBASE-KR8 PMD

Associated clause	800GBASE-KR8
170—RS	Required
170—800GMII ^a	Optional
171—800GMII Extender	Optional
172—PCS for 800GBASE-R	Required
173—PMA for 800GBASE-R	Required
120F—800GAUI-8 C2C	Optional
73—Auto-Negotiation	Required

^a The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

Insert the following paragraph after the fifth paragraph of 163.1:

For the 800GBASE-KR8 PHY, in order to support the required frame loss ratio (see [1.4.344](#)) of less than 6.2×10^{-11} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 3.4×10^{-12} after processing by the PCS (Clause 172) if there are negligible errors due to other electrical interfaces (800GAUI-n). If the PMD

and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 3.4×10^{-12} .

Replace Figure 163–1 with the following figure:

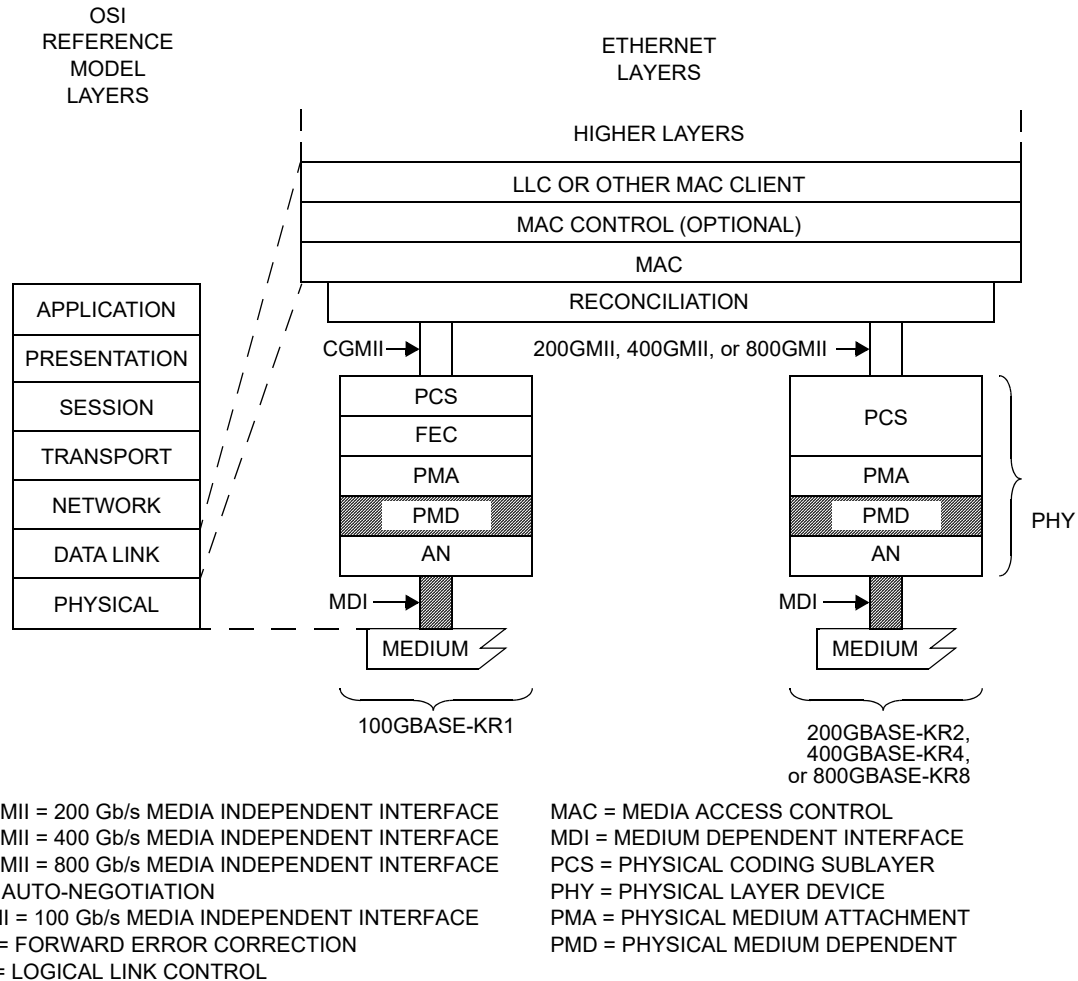


Figure 163–1—100GBASE-KR1, 200GBASE-KR2, 400GBASE-KR4, and 800GBASE-KR8 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

163.2 Conventions

Change 163.2 as follows:

Clause 163 describes ~~three-four~~ PMDs, 100GBASE-KR1, 200GBASE-KR2, ~~and~~ 400GBASE-KR4, ~~and~~ 800GBASE-KR8, which have one, two, ~~and four~~, ~~and eight~~ lanes, respectively. For efficient description, the parameter n is used to describe the number of lanes in a specific PMD. Accordingly, $n = 1$ for 100GBASE-KR1, $n = 2$ for 200GBASE-KR2, ~~and~~ $n = 4$ for 400GBASE-KR4, ~~and~~ $n = 8$ for 800GBASE-KR8.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n - 1$.

Within this clause, the unqualified term “PMD” refers to any of 100GBASE-KR1 PMD, 200GBASE-KR2 PMD, ~~or 400GBASE-KR4 PMD, or 800GBASE-KR8 PMD.~~

Within this clause, the term FEC refers to either [Clause 91](#) RS-FEC or ~~the~~ [Clause 161](#) RS-FEC-Int for 100GBASE-KR1, ~~or the RS-FEC within the~~ [Clause 119](#) PCS for 200GBASE-KR2 and 400GBASE-KR4, ~~or the RS-FEC within the~~ [Clause 172](#) PCS for 800GBASE-KR8.

163.3 PMD service interfaces

Change 163.3 as follows:

The service interfaces of the 100GBASE-KR1, 200GBASE-KR2, ~~and 400GBASE-KR4, and 800GBASE-KR8~~ PMDs are identical to those of the 100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4, and 800GBASE-CR8~~ PMDs, respectively (see 162.3).

163.5 Delay constraints

Change Table 163–4 as follows:

Table 163–4—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
100GBASE-KR1	4 096	8	40.96
200GBASE-KR2	8 192	16	40.96
400GBASE-KR4	16 384	32	40.96
<u>800GBASE-KR8</u>	<u>32 768</u>	<u>64</u>	<u>40.96</u>

^a One bit time is equal to 10 ps for 100GBASE-KR1, 5 ps for 200GBASE-KR2, ~~and 2.5 ps for 400GBASE-KR4, and 1.25 ps for 800GBASE-KR8.~~ (See [1.4.215](#) for the definition of bit time.)

^b One pause_quantum is equal to 5.12 ns for 100GBASE-KR1, 2.56 ns for 200GBASE-KR2, ~~and 1.28 ns for 400GBASE-KR4, and 0.64 ns for 800GBASE-KR8.~~ (See [31B.2](#) for the definition of pause_quanta.)

Change the second paragraph of 163.5 as follows:

Descriptions of overall system delay constraints can be found in [80.4](#) for 100GBASE-KR1 ~~and~~, in [116.4](#) for 200GBASE-KR2 and 400GBASE-KR4, ~~and in~~ [169.4](#) for 800GBASE-KR8.

163.6 Skew constraints

Insert new subclause 163.6.3 after 163.6.2 as follows:

163.6.3 Skew Constraints for 800GBASE-KR8

Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5.

163.8 PMD functional specifications

163.8.1 Link block diagram

Change the text in 163.8.1 as follows:

One direction of a 100GBASE-KR1, 200GBASE-KR2, ~~or 400GBASE-KR4~~, or 800GBASE-KR8 link is shown in Figure 163–2. The PMDs on both ends of the link have connected ground references.

Change the title of Figure 163–2 as follows:

Figure 163–2—100GBASE-KR1, 200GBASE-KR2, ~~or 400GBASE-KR4~~, or 800GBASE-KR8 link

163.9 Electrical characteristics

163.9.2 Transmitter characteristics

Change footnote a in Table 163–5 as follows (unchanged rows and footnotes not shown):

Table 163–5—Summary of transmitter specifications at TP0v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		53.125 ± 50 ppm ^a	GBd
...			

^a For a 100GBASE-KR1, 200GBASE-KR2, or 400GBASE-KR4 PMD in the same package as the PCS sublayer ~~or for any 800GBASE-KR8 PMD~~. In other cases, the signaling rate is derived from the input to the PMD transmit function provided by the adjacent PMA sublayer.

163.9.3 Receiver characteristics

Change Table 163-8 as follows (most unchanged rows not shown):

Table 163–8—Summary of receiver specifications at TP5v

Parameter	Reference	Value	Units
Signaling rate, each lane (range) <u>100GBASE-KR1, 200GBASE-KR2, or 400GBASE-KR4</u> <u>800GBASE-KR8</u>	163.9.3.1	53.125 ± 100 ppm <u>53.125 ± 50 ppm</u>	GBd <u>GBd</u>
...			
Jitter tolerance	163.9.3.6	Table 162–17	—
NOTE—For <u>100GBASE-KR1, 200GBASE-KR2, or 400GBASE-KR4</u> , although the PMD transmitter is specified with a signaling rate range of ±50 ppm when in the same package as the PCS, the signaling rate range may be ±100 ppm when derived from an intermediate interface (e.g., 100GAUI-4).			

163.9.3.1 Receiver signaling rate

Change 163.9.3.1 as follows:

A PHY shall comply with the receiver requirements of 163.9.3.5 and 163.9.3.6 for any signaling rate in the range specified in Table 163–8.

NOTE—Although the PMD transmitter is specified with a signaling rate range of ±50 ppm when in the same package as the PCS sublayer, the signaling rate range may be ±100 ppm when derived from an intermediate interface (e.g., 100GAUI-4).

163.9.3.5 Receiver interference tolerance

Change item i) in the lettered list in 163.9.3.5 as follows:

- i) The test pattern is the scrambled idle test pattern defined in 119.2.4.9 or 172.2.4.11.

163.10 Channel characteristics

Change 163.10 as follows:

Table 163–10 provides a summary of the channel characteristics for 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, and references to the subclauses addressing each parameter.

163.11 MDI specifications

The MDI for the 100GBASE-KR1, 200GBASE-KR2, ~~and 400GBASE-KR4~~, and 800GBASE-KR8 PHYs is an implementation-dependent direct electrical connection between the PMD and the medium. The MDI is composed of $2 \times n$ differential pairs, one pair for the transmit function and one pair for the receive function on each lane, marked by TP0 and TP5 in Figure 163–2.

Change the title of 163.13 as follows:

**163.13 Protocol implementation conformance statement (PICS) proforma for
Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium,
type 100GBASE-KR1, 200GBASE-KR2, ~~and 400GBASE-KR4~~, and 800GBASE-KR8⁹**

163.13.1 Introduction

Change the first paragraph of 163.13.1 as follows:

The supplier of a protocol implementation that is claimed to conform to Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, ~~and 400GBASE-KR4~~, and 800GBASE-KR8, shall complete the following protocol implementation conformance statement (PICS) proforma.

163.13.2 Identification

163.13.2.2 Protocol summary

Change the table in 162.13.2.2 as follows:

Identification of protocol standard	IEEE Std 802.3df-2024, Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 , <u>and 800GBASE-KR8</u>
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-2024.)	

Date of Statement	
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⁹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

163.13.3 Major capabilities/options

Change the table in 163.13.3 as follows (some unchanged rows not shown):

Item ^a	Feature	Subclause	Value/Comment	Status	Support
...					
*KR4	400GBASE-KR4 PMD	163.1	Can operate as a 400GBASE-KR4 PMD	O.1	Yes [] No []
*KR8	800GBASE-KR8 PMD	163.1	Can operate as an 800GBASE-KR8 PMD	O.1	Yes [] No []
PMA100	100GBASE-R PMA	163.1	Device implements Clause 135 PMA for 100GBASE-P	KR1:M	Yes [] N/A []
...					
PMA400	400GBASE-R PMA	163.1	Device implements Clause 120 PMA for 400GBASE-R	KR4:M	Yes [] N/A []
PMA800	800GBASE-R PMA	163.1	Device implements Clause 173 PMA for 800GBASE-R	KR8:M	Yes [] N/A []
*AUIPMD	100GAUI-n C2C, 200GAUI-n C2C, or 400GAUI-n C2C, or 800GAUI-n C2C	163.1	Service interface of PMA adjacent to PMD is physically instantiated	O.2	Yes [] No []
AUIFEC	CAUI-n C2C or 100GAUI-n C2C	163.1	Service interface of PMA between PCS and RS-FEC is physically instantiated	KR4:O.2	Yes [] No [] N/A []
*nGMII	CGMII, 200GMII, or 400GMII, or 800GMII, and RS	163.1	Interface is implemented or functionally equivalent	O.2	Yes [] No []
...					
PCS400	400GBASE-R PCS	163.1	Device implements Clause 119 400GBASE-R PCS	KR4* nGMII:M	Yes [] N/A []
PCS800	800GBASE-R PCS	163.1	Device implements Clause 172 800GBASE-R PCS	KR8* nGMII:M	Yes [] N/A []
PCSAN	PCS requirements for AN service interface	163.4	PCS supports service interface primitive AN_LINK.indication	nGMII:M	Yes [] N/A []
...					

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

Change the title of 163.13.4 as follows:

**163.13.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and
baseband medium, type 100GBASE-KR1, 200GBASE-KR2, ~~and 400GBASE-KR4, and~~
800GBASE-KR8**

163.13.4.2 PMD control function

*Change the rows for items PC2 and PC3 in the table in 163.13.4.2 as follows (unchanged rows not
shown):*

Item	Feature	Subclause	Value/Comment	Status	Support
...					
PC2	Training pattern	136.8.11.1.3 <u>162.8.11.1</u>	Each lane implements four generator polynomials defined in Table 136–8 Table 162–10a	M	Yes []
PC3	Training pattern	136.8.11.1.3 <u>162.8.11.1</u>	State set to the value of seed_ <i>i</i> at the start of the training pattern	M	Yes []
...					